

5 HCI Guidelines and Evaluation Techniques

5.1 User Interface Management System (UIMS)

Q.1 Explain Seeheim model of the logical components of user interface management system.

Ans.: The logical components of a UIMS were identified as :

1. **Presentation :** The component responsible for the appearance of the interface, including what output and input is available to the user.
2. **Dialog control :** The component which regulates the communication between the presentation and the application.
3. **Application interface :** The view of the application semantics that is provided as the interface.

• Fig. Q.1.1 shows Seeheim model of the logical components.

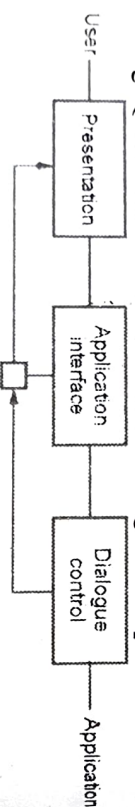


Fig. Q.1.1 : Seeheim model of the logical components

- The Seeheim workshop proposed a new model, which can be characterized by two principles :
1. The application program shall be separated from the user interface;
 2. The user interface shall be decomposed into three components: the presentation component, the dialogue component and the application interface component.

- The three components of the Seeheim model were interpreted for some time as the lexical, syntactic and semantic components of a language analyzer.
- In effect, the presentation handles low-level aspects comparable to tokens, while the dialogue component ensures that proper commands are specified by a set of tokens and the application interface describes the semantics of the application.

• This interpretation has been the motivation for using language description techniques to build user interfaces : we will describe a number of UIMSs which use such techniques.

Q.2 Explain advantages of constructing user interfaces with a UIMS.

Ans. :

1. UIMS improves the efficiency with which user interface designers can use their skills.
2. UIMS speeds the incremental process of creating a user interface.
3. UIMS makes it possible to create prototypes that can be discussed with the end user.
4. UIMS can adapt to different user profiles.
5. UIMS makes the integration of new application functionalities easier.
6. UIMS allows the application to be portable, while the user interface can be tailored to the particular environment.
7. UIMS can ease the debugging of interactive applications.

Q.3 List and explain various implementation techniques used for dialog modeling in UIMS.

Ans. :

1. **Menu networks :** The communication between application and presentation is modeled as a network of menus and submenus. To control the dialog, the programmer must simply encode the levels of menus and the connections between one menu and the next submenu or an action.

2. Grammar notations : The dialog between application and presentation can be treated as a grammar of actions and responses and, therefore, described by means of a formal context-free grammar notation.

3. State transition diagrams : State transition diagrams can be used as a graphical means of expressing dialog.

4. Event languages : Event languages are similar to grammar notations, except that they can be modified to express directionality and support some semantic feedback.

5. Constraints : Constraints systems are a special subset of declarative languages. Constraints can be used to make explicit the connection between independent information of the presentation and the application.

5.2 Goals of Evaluation

Q.4 What is evaluation ?

Ans. :

- It is needed to assess the designs and test systems to ensure that they actually behave as user expects and meet user requirements. This is the role of evaluation.

- Evaluation should not be thought of as a single phase in the design process. Ideally, evaluation should occur throughout the design life cycle, with the results of the evaluation feeding back into modifications to the design.

Q.5 List the factors that distinguishing evaluation techniques.

Ans. :

- There are eight factors that distinguish different evaluation techniques and therefore help us to make an appropriate choice.
- These are :

1. The stage in the cycle at which the evaluation is carried out
2. The style of evaluation

3. The level of subjectivity or objectivity of the technique
4. The type of measures provided
5. The information provided
6. The immediacy of the response
7. The level of interference implied
8. The resources required.

Q.6 Explain goals of evaluation.

Ans. : Main goals of evaluation are as follows :

1. **To assess the extent and accessibility of the system's functionality :** This includes not only making the correct functionality available within the system, but making it clearly reachable by the user in terms of the actions that the user needs to take to carry out the task. It also includes matching the use of the system to the user's expectations of the task.

2. **To assess users' experience of the interaction :** This includes considering features such as how easy the system is to understand, its usability and the user's satisfaction with it. It may also include his enjoyment and emotional response, particularly in the case of systems that are aimed at leisure or entertainment.

3. **To identify any specific problems with the system :** This is correlated to both the functionality and usability of the design. It is specifically concerned with identifying trouble-spots which can then be rectified.

5.3 Categorization of Evaluation Techniques

Q.7 Explain various category of evaluation techniques.

Ans. :

- Depending on the setting, user participation and control level, evaluations is classified into three large categories. They are as follows :

a) **Controlled user settings** : The activity of the users is controlled by hypotheses and measurements or compliance. Usability tests and experiments are the main methods. Example : usability labs and research laboratories

b) **Natural user settings** : User activities can be determined little or no control how in the real world the product would be used. Field studies are the main method used in-wild studies, for instance. Examples include online communities and products used in public places.

c) **Any settings that do not include users directly** : The most obvious usability problems are criticized by consultants and investigators for, predicted and model aspects of the interface. Methods cover inspections, heuristics, pathways, models and analysis.

Q.8 Design an experiment to test whether adding color coding to an interface will improve accuracy. Identify your hypothesis, participant group, dependent and independent variables, experimental design, task and analysis approach.

Ans [SPPU : Dec-17, May-18, End Sem, Marks 8]

Ans : The following is only an example of the type of experiment that might be devised.

1. **Participants** : Taken from user population.
2. **Hypothesis** : Color coding will make selection more accurate.
3. **IV (Independent Variable)** : Color coding.
4. **DV (Dependent Variable)** : Accuracy measured as number of errors.

5. **Design** : Between-groups to ensure no transfer of learning (or within-groups with appropriate safeguards if participants are scarce).

6. **Task** : The interfaces are identical in each of the conditions, except that, in the second, color is added to indicate related menu items. Participants are presented with a screen of menu choices

(ordered randomly) and verbally told what they have to select. Selection must be done within a strict time limit when the screen clears. Failure to select the correct item is deemed an error. Each presentation places items in new positions. Participants perform in one of the two conditions.

7. **Analysis** : t test.

5.4 DECIDE

Q.9 What is DECIDE evaluation framework ?

Ans : DECIDE is a framework that is used to guide evaluation.

1. Determine the goals the evaluation addresses.
2. Explore the specific questions to be answered.
3. Choose the evaluation paradigm and techniques to answer the questions.
4. Identify the practical issues.
5. Decide how to deal with the ethical issues.
6. Evaluate, interpret and present the data.

Q.10 Explain various styles of evaluation.

Ans :

1. **Laboratory studies** : Performed under laboratory conditions. Laboratory studies do not have the overhead of installing or updating a system in a real work site, so they permit rapid cycles of user feedback and prototyping.

Advantages :

- a. Specialist equipment available
- b. Uninterrupted environment

Disadvantages :

- a. Lack of context
- b. Difficult to observe several users cooperating

2. Field studies : Conducted in the work environment or in "the field". Field studies are done in natural settings. The aim is to understand what users do naturally and how technology impacts them.

Advantages :

- Natural environment
- Context retained
- Longitudinal studies possible

Disadvantages :

- Distractions, interruptions
- Noise

5.5 Heuristic Evaluation

Q.11 Explain Nielsen's ten heuristics.

[SPPU : Dec-17, May-18, End Sem, Marks 8]

Ans. : Nielsen's ten heuristics are :

- 1. Visibility of system status :** The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.
- 2. Match between system and the real world :** The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.
- 3. User control and freedom :** Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.
- 4. Consistency and standards :** Users should not have to wonder whether different words, situations or actions mean the same thing. Follow platform conventions.

5. Error prevention : Even better than good error messages is careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.

6. Recognition rather than recall : Minimize the user's memory load by making objects, actions and options visible. The user should not have to remember information from one part of the dialogue to another.

7. Flexibility and efficiency of use : Accelerators unseen by the novice user, may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

8. Aesthetic and minimalist design : Dialogues should not contain information which is irrelevant or rarely needed.

9. Help users recognize, diagnose and recover from errors : Error messages should be expressed in plain language, precisely indicate the problem and constructively suggest a solution.

10. Help and documentation : Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation.

Q.12 Discuss Heuristic evaluation.

Ans. :

- A third expert-based approach is the use of models. Certain cognitive and design models provide a means of combining design specification and evaluation into the same framework.
- For example, the GOMS (goals, operators, methods and selection) model predicts user performance with a particular interface and can be used to filter particular design options. Similarly, lower-level modeling techniques such as the keystroke-level model provide predictions of the time users will take to perform low-level physical tasks.

- Design methodologies, such as design rationale, also have a role to play in assessment at the design stage. Design rationale provides a framework in which design options can be evaluated.
- Dialog models can also be used to estimate dialog sequences for problems, such as unreachable states, circular dialogs and complexity. Models such as state transition networks are useful for evaluating dialog designs prior to implementation.

5.6 Cognitive Walkthrough

Q.13 The cognitive walkthrough is a formalized way of imagining people's thoughts and actions when they use an interface for the first time. During a cognitive walkthrough the evaluator needs to ask four questions as below.

- i) Is the effect of the action the same as the user's goal at that point?
 - ii) Will users see that the action is available?
 - iii) Once users have found the correct action, will they know it is the one they need?
 - iv) After the action is taken, will users understand the feedback they get?
- Given below is an action sequence for creating a customized voicemail message on an iPhone.

- 1) Tap Voicemail
- 2) Tap Greeting
- 3) Tap Custom
- 4) Tap Record and speak your greeting.
- 5) When you finish, tap Stop.
- 6) To listen to your greeting, tap Play.
- 7) To re-record, repeat steps 4 and 5.
- 8) Tap Save.

Imagine an iPhone interface and create a report of the cognitive walkthrough for the above mentioned task in context with the review questions.

Ans. :

 [SPPU : May-18, End Sem, Marks 8]

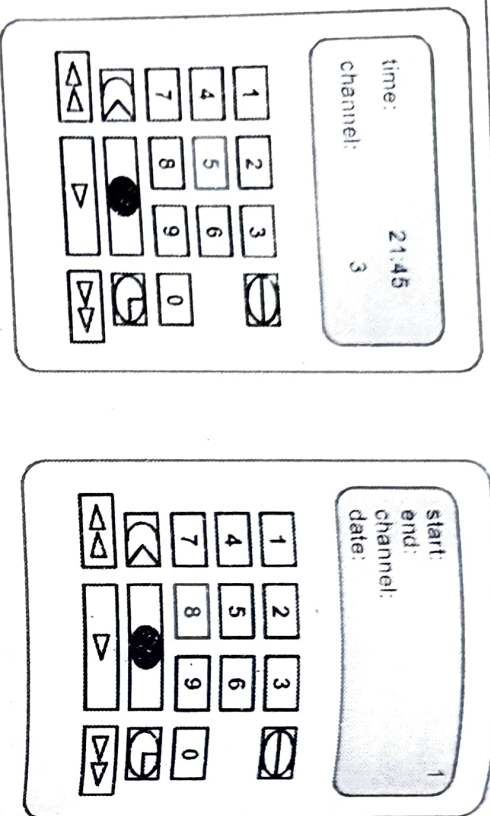
- Formalized way of imagining people's thoughts and actions when they use an interface for the first time.
- First select a task that the design is intended to support.

- Then try to tell a believable story about each action a user has to take to do the task.
- To make the story believable, you have to motivate each of the user's actions, relying on the user's general knowledge and on the prompts and feedback provided by the interface. If you can't tell a believable story about an action, then you've located a problem with the interface.
- Question assumptions about what the users will be thinking
- Identify controls that may be missing or hard to find
- Note inadequate feedback.
- Suggest difficulties with labels and prompts.
- Vocabulary Problem : On a piece of paper write the name you would give to a program that tells about interesting activities occurring in some major metropolitan area.
- Focus most clearly on problems that users will have when they first use an interface, without training.
- Not a technique for evaluating the system over time (e.g., how quickly a user moves from beginner to intermediate).
- Most effective if designers can really create a mental picture of the actual environment of use.
- Prior to doing a walkthrough, you need four things :
 1. You need a description of a prototype of the interface. It doesn't have to be complete, but it should be fairly detailed. Things like exactly what words are in a menu can make a big difference.
 2. You need a task description (for a representative task).
 3. You need a complete, written list of the actions needed to complete the task.
 4. You need an idea of who the users will be and what kind of experience they'll bring to the job.

Q.14 Explain cognitive walkthrough.**Ans. :**

- Heuristic evaluation can be carried out on a design specification so it is useful for evaluating early design. But it can also be used on prototypes, storyboards and fully running systems. It is a flexible, relatively cheap approach. Hence it is often considered a *discount usability* technique.
 - The general idea behind heuristic evaluation is that several evaluators independently review a system to come up with probable usability problems.
 - Each evaluator assesses the system and notes violations of any of heuristics that would indicate a probable usability problem.
 - The evaluator also assesses the severity of each usability problem, based on four factors : how common is the problem, how easy is it for the user to overcome, will it be a one-off problem or a persistent one and how seriously will the problem be perceived ? These can be combined into an overall severity rating on a scale of 0-4 :
 - 0 = I don't agree that this is a usability problem at all
 - 1 = Cosmetic problem only : need not be fixed unless extra time is available on project
 - 2 = Minor usability problem : fixing this should be given low priority
 - 3 = Major usability problem : important to fix, so should be given high priority
 - 4 = Usability catastrophe : imperative to fix this before product can be released (Nielsen)
 - Nielsen recommends the use of ten heuristics given below as providing the most effective coverage of the most common usability problems.
1. **Visibility of system status** : Always keep users informed about what is going on, through appropriate feedback within reasonable time.

2. **Match between system and the real world** : The system should speak the user's language, with words, phrases and concepts familiar to the user, rather than system-oriented terms.
 3. **User control and freedom** : Users often choose system functions by mistake and need a clearly marked 'emergency exit' to leave the unwanted state without having to go through an extended dialog. Support undo and redo.
 4. **Consistency and standards** : Users should not have to wonder whether words, situations or actions mean the same thing in different contexts.
 5. **Error prevention** : Make it difficult to make errors.
 6. **Recognition rather than recall** : Make objects, actions and options visible. The user should not have to remember information from one part of the dialog to another.
 7. **Flexibility and efficiency of use** : Allow users to tailor frequent actions.
 8. **Aesthetic and minimalist design** : Dialogs should not contain information that is irrelevant or rarely needed.
 9. **Help users recognize, diagnose and recover from errors** : Error messages should be expressed in plain language (no codes), precisely indicate the problem and constructively suggest a solution.
 10. **Help and documentation** : Few systems can be used with no instructions so it may be necessary to provide help and documentation.
- Q.15 Complete the cognitive walk-through example for the video remote control design.**
- Ans. :**
- ISPPU : Dec-19, End Sem, Marks 9]**
- Cognitive walkthrough is a detailed review of a sequence of actions, in this case, the steps that an interface will require the user to perform in order to accomplish some known task.



- The evaluators go through each step and provide a story about why that step is not good for new users.
- To do so, you need four things : a prototype of the system, a description of the task the user is to perform on the system, a complete written list of the actions needed to complete the task with the system and an indication of who the users are and what kind of experience and knowledge the evaluators can assume about them.

- In terms of the user's action (UA) and the system's display or response (SD). The initial display is as the left-hand picture in Figure.

- UA 1 : Press the 'timed record' button
- SD 1 : Display moves to timer mode. Flashing cursor appears after 'start:'
- UA 2 : Press digits 1 8 0 0
- SD 2 : Each digit is displayed as typed and flashing cursor moves to next position
- UA 3 : Press the 'timed record' button
- SD 3 : Flashing cursor moves to 'end:'
- UA 4 : Press digits 1 9 1 5

- SD 4 : Each digit is displayed as typed and flashing cursor moves to next position
- UA 5 : Press the 'timed record' button
- SD 5 : Flashing cursor moves to 'channel:'
- UA 6 : Press digit 4
- SD 6 : Digit is displayed as typed and flashing cursor moves to next position
- UA 7 : Press the 'timed record' button
- SD 7 : Flashing cursor moves to 'date:'
- UA 8 : Press digits 2 4 0 2 0 5
- SD 8 : Each digit is displayed as typed and flashing cursor moves to next position
- UA 9 : Press the 'timed record' button
- SD 9 : Stream number in top right-hand corner of display flashes
- UA 10 : Press the 'transmit' button
- SD 10 : Details are transmitted to video player and display returns to normal mode
- Having determined our action list we are in a position to proceed with the walkthrough. For each action we must answer the four questions and tell a story about the usability of the system.
- Beginning with UA 1 :
- UA 1 : Press the 'timed record' button
- Question 1 : Is the effect of the action the same as the user's goal at that point ?
- The timed record button initiates timer programming. It is reasonable to assume that a user familiar with VCRs would be trying to do this as his first goal.
- Question 2 : Will users see that the action is available ?
- The 'timed record' button is visible on the remote control.

- Question 3 : Once users have found the correct action, will they know it is the one they need ?
- It is not clear which button is the 'timed record' button. The icon of a clock is a possible candidate but this could be interpreted as a button to change the time. Other possible candidates might be the fourth button down on the left or the filled circle. In fact, the icon of the clock is the correct choice but it is quite possible that the user would fail at this point. This identifies a potential usability problem.
- Question 4 : After the action is taken, will users understand the feedback they get ?
- Once the action is taken the display changes to the timed record mode and shows familiar headings (start, end, channel, date). It is reasonable to assume that the user would recognize these as indicating successful completion of the first action.

5.7 Usability Testing

Q.16 Write short note on usability testing.

Ans. :

- It is a testing of product by the actual user. It is to improve usability of product. It shows how users are doing their tasks using this product. Testers can observe as well as record the participants, can analyze the data obtained and to the changes accordingly.
- This usability testing focuses on requirements, empirical measurement and iterative design.
- Many iterations are performed before final product release in public. It is just like rehearsals for any play. Previously this usability testing was performed in laboratory by the experts which were good in user interface design and testing.
- This laboratory requires more cost so replaced by mobile usability testing kits.

- Another many techniques for usability testings are think aloud technique. In this technique user is asked to perform all the steps of actions. In videotaping, designers can get true review about the participants and problems in the design.
- In interviews and user satisfaction questionnaires, designer enable to evaluate the users.
- Through usability testing designer can get more information about time required to learn a specific function, speed of task performance, types and rate of errors, user retention of commands, over time, subjective user satisfaction.
- Even this testing is continued after product get released. After release, testing through the interviews and focus groups. This gives an important information about problems with functionality and user satisfaction. Data logs can be performed the testing.
- This usability testing is having same limitations as it is not covering all features of interface. As it is testing in laboratory only rather than actual environment.

END ... ✍